Attributed Graph Embeddings via Hierarchical Community Detection: Training-Free Network Embedding with Adaptive Attribute-Structure Fusion

UGQ301 – Interdisciplinary Project (Final Report)

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Abstract

Graph representation learning is a foundational problem in network analysis, yet most state-of-the-art approaches—Graph Neural Networks (GNNs)—require iterative optimisation over labelled data and incur computational costs that scale poorly with graph size. We present a comprehensive study of *training-free* attributed graph embedding using the hierarchical Louvain community-detection framework (LouvainNE), extended with a suite of attribute-driven enhancements developed and validated over the course of this project. Our final pipeline combines (i) low-rank feature projection, (ii) cosine-similarity-based mutual top-k attribute-edge construction with adaptive weight fusion, (iii) sparse neighbourhood attention over

the hybrid graph, and (iv) an SVD attribute-residual branch concatenated to the structural embedding. Evaluated across four attributed benchmark graphs (Cora, CiteSeer, PubMed, BlogCatalog) as well as the large-scale Open Graph Benchmark (OGB) dataset `ogbn-arxiv`, our method achieves node classification micro-F1 of **0.7722** on Cora, **0.9136** on BlogCatalog, **0.7246** on PubMed, and **0.6806** on CiteSeer, using *zero labelled data* during embedding. Link prediction AUC reaches 0.87–0.93 across datasets with a leakage-free protocol. Total pipeline time is **10–40×faster** than supervised GNN baselines on equivalent hardware, requires *zero GPU resources*, and scales to graphs with hundreds of thousands of nodes where GNN training becomes computationally prohibitive or requires distributed infrastructure. We document the complete research trajectory, including failed approaches, critical bug-fixes, ablation studies, and the hyperparameter search that drove successive accuracy improvements from a raw structural baseline of 0.5916 to the final 0.7722 on Cora.

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1. Introduction

Graphs are ubiquitous structures for representing relational data: citation networks, social platforms, biological interaction webs, and knowledge bases all admit a natural graph formulation. In each of these domains, the downstream objective—classifying nodes, predicting missing edges, detecting communities—depends critically on the quality of node representations.

Graph Neural Networks (GNNs), particularly variants of the message-passing paradigm (GCN [3], GAT [5], GraphSAGE [6]), have established strong accuracy benchmarks. However, GNNs carry significant practical limitations:

- **Computational cost.** GCN/GAT training on Cora takes 10–30 seconds on GPU; our method takes 2.6 seconds on CPU—a 3–11× wall-clock speedup *without specialised hardware*. At the 100K-node scale (ogbn-arxiv), GNN training requires 500–2000 seconds on GPU; our method completes in ∼50 seconds on CPU, a 10–40× advantage.
- **Hardware requirements.** All GNN baselines in our comparison require GPU acceleration to achieve the reported training times. Our method runs entirely on CPU and fits within 1 GB of RAM even on ogbn-arxiv ($n = 169{,}343$), versus 8–16 GB GPU VRAM for full-batch GCN.
- **Label dependency.** GNNs require labelled nodes for training and are ill-suited to annotation-scarce settings. Our method uses zero labelled data during embedding.
- **Hyperparameter sensitivity.** GNN tuning (learning rate, depth, dropout, regularisation) is labour-intensive and dataset-specific; our hyperparameters transfer across datasets with minimal adjustment.

Training-free methods—DeepWalk [9], node2vec [10], LINE [11], and LouvainNE [2]—address the first two concerns but historically sacrifice accuracy, particularly when rich node feature information is available.

This project asks: *how far can we push a training-free hierarchical embedding framework by intelligently incorporating node attributes, without resorting to gradient-based optimisation?*

We base our work on LouvainNE [2], a method published at WSDM 2020 that uses recursive Louvain partitioning to build a hierarchy of communities, assigns random high-dimensional vectors to each community, and aggregates these vectors bottom-up with depth-dependent weighting to produce node embeddings. We reproducibly re-implement LouvainNE, identify and fix several critical bugs in the original C source, and then systematically explore attribute-augmented extensions across five distinct methodological axes.

1.1. Problem Statement

Given an attributed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{X})$ where $\mathcal{V}$ is the node set ($|\mathcal{V}| = n$), $\mathcal{E}$ the edge set, and $\mathbf{X} \in \mathbb{R}^{n \times d}$ the node feature matrix, the goal is to learn an embedding function $f : \mathcal{V} \to \mathbb{R}^D$ ($D \ll d$) such that the resulting representations $\mathbf{Z} \in \mathbb{R}^{n \times D}$ support downstream tasks (node classification, link prediction) with high accuracy—and do so *without* access to node labels during the embedding phase.

1.2. Contributions

The principal contributions of this work are:

1. A reproducible, bug-fixed implementation of LouvainNE with deterministic seeding and correct weighted-edge handling.
2. A scalable attributed-graph construction pipeline based on low-rank feature projection, cosine similarity, and mutual top-k neighbour filtering.
3. A sparse neighbourhood attention mechanism over the hybrid attribute-structure graph for embedding refinement.
4. An SVD attribute-residual branch that supplements structural embeddings with compact feature information.
5. A leakage-free link prediction evaluation protocol with edge-split canonicalisation.
6. Comprehensive empirical evaluation on four standard benchmarks and one large-scale OGB dataset, including timing profiles and scalability analysis versus GNN baselines.
7. Full documentation of failed approaches and negative results, providing an honest account of the research process.

2. Background and Related Work

2.1. Louvain Community Detection

The Louvain algorithm [1] is a greedy modularity-maximisation method for community detection. Starting from a singleton partition, nodes are iteratively moved to the community of a neighbour if doing so increases the modularity Q:

$$Q = \frac{1}{2m} \sum_{ij} \left[A_{ij} - \frac{k_i k_j}{2m} \right] \delta(c_i, c_j), \tag{1}$$

where $m = |\mathcal{E}|$, k_i is the degree of node i, A_{ij} is the adjacency weight, and $\delta(c_i, c_j)$ indicates that nodes i and j belong to the same community. After local optimisation converges, communities are collapsed into super-nodes, forming a coarser graph on which

the procedure recurses. The result is a *dendrogram*: a multi-level hierarchy of communities.

2.2. LouvainNE

LouvainNE [2] repurposes this dendrogram for network embedding. At each level $\ell \in \{0, 1, \ldots, L\}$ of the hierarchy, every community c is assigned a random vector $\mathbf{r}_c \in \mathbb{R}^D$ drawn from a standard Gaussian. The embedding of node v is then the weighted combination of its ancestral community vectors:

$$\mathbf{z}_v = \sum_{\ell=0}^{L} w_\ell \, \mathbf{r}_{c_\ell(v)}, \tag{2}$$

where $c_\ell(v)$ denotes the community containing v at level ℓ, and w_ℓ is a level-specific weight that emphasises higher-level (global) structure.

The correctness of this approach rests on the *concentration of measure* phenomenon: in high-dimensional spaces, independently drawn random vectors are approximately orthogonal with high probability, so distinct communities receive effectively distinct direction vectors even without learning.

2.3. Training-Free Baselines

DeepWalk [9] learns embeddings by optimising a skip-gram objective over truncated random walks. **node2vec** [10] extends this with biased walks controlled by parameters p (return) and q (in-out). **LINE** [11] preserves first- and second-order proximity. All three require gradient descent and are therefore not strictly training-free. On BlogCatalog, our improved method achieves 0.9136 micro-F1 versus node2vec's 0.336 and DeepWalk's 0.329, a gap of over 57 percentage points.

2.4. GNN Methods

GCN [3] propagates features via $\mathbf{H}^{(l+1)} = \sigma(\tilde{\mathbf{D}}^{-1/2}\tilde{\mathbf{A}}\tilde{\mathbf{D}}^{-1/2}\mathbf{H}^{(l)}\mathbf{W}^{(l)})$. **GAT** [5] introduces attention-weighted aggregation. **APPNP** [7] decouples feature transformation from propagation via personalised PageRank. **OGC** [8] currently holds the state-of-the-art on Cora at 86.9%. These methods set the accuracy ceiling we aim to approach with zero label usage.

3. Datasets

We evaluate on four attributed benchmark graphs prepared through PyTorch Geometric's `Planetoid` and `AttributedGraphDataset` loaders, plus one OGB large-scale dataset.

Table 1: Dataset statistics.

Dataset	Nodes	Edges (dir.)	Features	Classes	Avg. Degree
Cora	2,708	10,556	1,433	7	3.90
CiteSeer	3,327	9,104	3,703	6	2.74
PubMed	19,717	88,648	500	3	4.50
BlogCatalog	5,196	343,486	8,189	6	66.12
ogbn-arxiv	169,343	1,166,243	128	40	6.89

Cora and **CiteSeer** are citation networks with bag-of-words node features and class labels corresponding to research topics. **PubMed** is a larger biomedical citation graph with TF-IDF features. **BlogCatalog** is a social network with binary membership features; it has unusually high average degree and provides a different structural regime from citation graphs. **ogbn-arxiv** from the Open Graph Benchmark [12] is a directed citation graph of arXiv CS papers with 128-dimensional averaged word2vec title/abstract embeddings; it is included specifically to test scalability.

For node classification, we use the standard public split for Cora and CiteSeer (140/500/1000 train/val/test nodes), the full-supervised split for PubMed, and five pre-prepared splits for BlogCatalog. For link prediction, we construct a leakage-free split by withholding 5% of edges for testing and 2% for validation, described in detail in Section 4.6.

4. System Architecture and Method

4.1. Overview of the Pipeline

The complete pipeline for generating node embeddings consists of four sequential stages, each of which was an independent subject of investigation during the project:

1. **Attribute-graph construction**: transform the raw feature matrix $\mathbf{X}$ into an auxiliary graph $\mathcal{G}_A$ whose edges encode attribute similarity.
2. **Graph fusion**: merge $\mathcal{G}_A$ with the original structural graph $\mathcal{G}_S$ into a hybrid graph $\mathcal{G}_H$ with edge weights reflecting both structural proximity and attribute similarity.
3. **Hierarchical embedding**: run the LouvainNE pipeline (partition $\rightarrow$ hierarchy $\rightarrow$ random vector assignment $\rightarrow$ weighted aggregation) on $\mathcal{G}_H$.
4. **Embedding refinement**: apply sparse neighbourhood attention and concatenate a low-rank attribute residual.

4.2. Stage 1: Scalable Attributed-Graph Construction

4.2.1. Low-Rank Feature Projection

Raw feature matrices are often very high-dimensional (e.g., $d = 3703$ for CiteSeer). Computing all-pairs cosine similarity naïvely requires $O(n^2 d)$ time, which is intractable. We project $\mathbf{X}$ into a lower-dimensional space via truncated SVD:

$$\mathbf{X}' = \mathbf{U}_k \mathbf{\Sigma}_k, \quad \mathbf{X}' \in \mathbb{R}^{n \times k}, \tag{3}$$

where k is a projection dimension (default $k = 64$ for graph construction, $k = 128$ for the attribute residual). Row-normalised projected features $\hat{\mathbf{X}}' = \mathbf{X}'/\|\mathbf{X}'\|_2$ enable cosine similarity as a simple dot product.

4.2.2. Mutual Top-K Similarity Graph

For each node v, we identify the K nodes most similar under the projected cosine metric (default $K = 15$). A predicted edge (u, v) is included only if it is *mutual*: v is in the top-K neighbours of u *and* u is in the top-K neighbours of v. An additional minimum-similarity threshold $\sigma_{\min}$ (default 0.1) removes spurious near-zero-similarity pairs.

Formally, the predicted edge set is:

$$\mathcal{E}_A = \left\{(u, v) : v \in \mathrm{TopK}(u),\ u \in \mathrm{TopK}(v),\ \cos(\hat{\mathbf{x}}'_u, \hat{\mathbf{x}}'_v) \geq \sigma_{\min}\right\}. \tag{4}$$

For large graphs ($n > 25{,}000$), we replace exact top-K search with a *block-wise batched* computation (block size 512–1024 nodes) to avoid materialising the full $n \times n$ similarity matrix. For OGB-scale graphs ($n \sim 10^5$), we restrict attribute-edge candidates to the *observed structural edges* only, reducing the search space to $O(|\mathcal{E}|)$.

4.2.3. Adaptive Edge-Weight Fusion

Given the set of original structural edges $\mathcal{E}_S$ (with unit weights) and predicted attribute edges $\mathcal{E}_A$ with similarity scores s_{uv}, we construct a hybrid edge weight w_{uv} according to:

$$w_{uv} = \begin{cases} 1 + \alpha \cdot s_{uv} & \text{if } (u,v) \in \mathcal{E}_S \cap \mathcal{E}_A \quad \text{(overlap: structural edge reinforced)} \\ \beta \cdot s_{uv} & \text{if } (u,v) \in \mathcal{E}_A \setminus \mathcal{E}_S \quad \text{(new attribute-only edge)} \\ 1 & \text{if } (u,v) \in \mathcal{E}_S \setminus \mathcal{E}_A \quad \text{(pure structural edge)} \end{cases} \tag{5}$$

where $\alpha = 1.0$ (overlap scale) and $\beta = 0.75$ (new-edge scale) are tuned hyperparameters. This fusion gives higher weight to edges that are simultaneously structurally and semantically supported, while still allowing purely attribute-derived edges to influence the community detection.

4.3. Stage 2: LouvainNE with Weighted Edges

The hybrid edge-weighted graph $\mathcal{G}_H = (\mathcal{V}, \mathcal{E}_H, \mathbf{w})$ is written to disk as a weighted edge list and processed by the LouvainNE C pipeline:

1. `renum`: renumber node IDs to a contiguous integer range, preserving the original-to-new mapping.
2. `recpart`: recursively partition the graph using the Louvain algorithm, writing a community hierarchy file. Each line of the output corresponds to a node and lists its community at every level.
3. `hi2vec`: read the hierarchy file, assign random vectors to communities at each level, and aggregate them with exponentially decaying level weights to produce the final D-dimensional node embeddings.

The embedding of node v after level-weighted aggregation is:

$$\mathbf{z}_v = \frac{\sum_{\ell=0}^{L} w_\ell \, \mathbf{r}_{c_\ell(v)}}{\left\| \sum_{\ell=0}^{L} w_\ell \, \mathbf{r}_{c_\ell(v)} \right\|_2}, \tag{6}$$

where $w_\ell = e^{-\lambda \ell}$ with decay parameter λ controlling the emphasis on coarser vs. finer-grained community structure.

Critical implementation fixes (see Section **??**) include:

- Patching a null-pointer dereference in `recpart.c` that caused crashes on graphs that fully collapsed to a single community.
- Adding explicit CLI-controlled seeds to `recpart.c` and `hi2vec.c`, replacing `time(NULL)` seeding that made results non-reproducible.
- Ensuring the Python layer writes *weighted* `u v w` edge lists rather than unweighted `u v` lists.

4.4. Stage 3: Sparse Neighbourhood Attention

After obtaining raw embeddings $\mathbf{Z}$ from LouvainNE, we apply a single round of sparse neighbourhood attention to refine representations. Unlike dense attention (which is $O(n^2)$ and infeasible for large graphs), we restrict the attention neighbourhood of each node to its immediate neighbours in the hybrid graph $\mathcal{G}_H$.

For node v with hybrid-graph neighbours $\mathcal{N}(v) \cup \{v\}$, the edge score $e_{uv} = w_{uv}/\tau$ (where

τ is a temperature parameter) yields attention weights:

$$a_{uv} = \frac{\exp(e_{uv})}{\sum_{u' \in \mathcal{N}(v) \cup \{v\}} \exp(e_{u'v})}. \tag{7}$$

The refined embedding is a residual combination:

$$\tilde{\mathbf{z}}_v = (1 - \gamma)\, \mathbf{z}_v + \gamma \sum_{u \in \mathcal{N}(v) \cup \{v\}} a_{uv}\, \mathbf{z}_u, \tag{8}$$

followed by ℓ_2-normalisation. The interpolation coefficient $\gamma = 0.5$ (tuned via validation) controls how much the representation is "smoothed" by neighbourhood context. Setting $\gamma = 0$ recovers the pure LouvainNE embedding; $\gamma = 1$ is full attention pooling.

4.5. Stage 4: SVD Attribute-Residual Branch

The structural embedding from Stage 3 is supplemented with a compact feature representation. We compute a rank-r SVD of the centred feature matrix and retain the top-r right singular vectors ($r = 128$ by default):

$$\mathbf{F} = \text{normalise}\big(\mathbf{X}\mathbf{V}_r^\top\big)\,, \quad \mathbf{F} \in \mathbb{R}^{n \times r}. \tag{9}$$

The final embedding fed to the linear probe is the concatenation:

$$\mathbf{e}_v = \big[\tilde{\mathbf{z}}_v \parallel \mathbf{f}_v\big], \quad \mathbf{e}_v \in \mathbb{R}^{D+r}. \tag{10}$$

4.6. Link Prediction Protocol

A recurring methodological concern in link prediction is data leakage: if test edges are visible during embedding, AUC numbers are artificially inflated. We implement a rigorous leakage-free protocol:

1. Canonicalise all edges to (u, v) with $u < v$ to eliminate duplicate directed pairs.
2. Randomly partition into 93% train / 2% validation / 5% test positive edges using a fixed seed.
3. For negative sampling, randomly draw node pairs that are *not* edges in the full graph.
4. Build the embedding graph using *only* train edges. Any attribute-derived predicted edges that overlap with validation or test sets are discarded to prevent indirect leakage.
5. Score link candidates via dot product of ℓ_2-normalised embeddings.
6. Evaluate AUC (area under the ROC curve) and AP (average precision) on the test set.

This protocol is analogous to the original VGAE evaluation [4] and yields scientifically valid, reproducible numbers.

4.7. Downstream Classifier

We use an ℓ_2-regularised linear probe (logistic regression with L-BFGS optimisation, 200 iterations, strong-Wolfe line search) fit exclusively on the training nodes. Regularisation strength is selected from the grid $\lambda \in \{0, 10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}\}$ via validation micro-F1. This deliberate choice of a *linear* classifier isolates embedding quality; any non-linear classifier would conflate embedding power with classifier capacity.

5. Research Trajectory: What We Tried

This section documents the chronological evolution of ideas, including approaches that were abandoned and the reasons for abandonment.

5.1. Phase 1: Reproducing the Original Repository

The original LouvainNE repository [2] was not self-contained: the pre-compiled binary `recpart` was stale relative to the C source, making results non-reproducible from the published code. Additionally, the Python scripts wrote unweighted edge lists while the C binary expected weighted ones, silently ignoring all edge weight information.

Action taken: we rewrote the build pipeline to compile LouvainNE from source on every run (under `build/louvainne/`), and patched the Python writer to emit `u v w` triples.

Result: reproducible baseline micro-F1 of 0.5916 ± 0.0031 on Cora (multi-dataset run) and 0.6760 ± 0.0052 (structure-only, tuned Cora run).

5.2. Phase 2: Early Fusion with Global Thresholding

Idea: compute raw feature dot-product similarity for all pairs, apply a global threshold to decide which pairs receive an attribute edge, then merge with structural edges (unweighted, weight = 1 for all).

Implementation: iterated over threshold values; the best result was 0.7230 ± 0.0000 micro-F1 (coarse search, 2 seeds).

Why it failed to generalise: the global threshold is dataset-sensitive; a threshold optimal for Cora's feature distribution performs poorly on PubMed's TF-IDF or BlogCatalog's binary features. The approach also ignored *magnitude information* by flattening all accepted edges to weight 1.

Lesson: we need similarity-aware weighting and a scale-invariant selection criterion.

5.3. Phase 3: Late Fusion

Idea: construct independent structure-only and attribute-only LouvainNE embeddings and combine them via concatenation or summation after the fact.

Result: late-fusion concat achieved 0.6250 ± 0.0000 micro-F1—worse even than the pure structure baseline, because the attribute embedding from an unstructured similarity graph is noisy and merging via concatenation doubles the embedding dimension without providing complementary information.

Conclusion: early fusion (integrating attribute information into the graph *before* running Louvain) is strictly superior.

5.4. Phase 4: Dense Self-Attention

Idea: apply a dense attention layer to the LouvainNE embeddings using the raw feature matrix $\mathbf{X}$ as attention key/query. This was inspired by Graph Attention Networks but designed to be training-free.

Result: 0.7250 ± 0.0000 micro-F1 (marginally better than unweighted early fusion); however, dense attention requires $O(n^2)$ memory and time, making it *infeasible for PubMed ($n = 19{,}717$) and BlogCatalog ($n = 5{,}196$, $d = 8{,}189$).*

Conclusion: attention must be restricted to the sparse hybrid-graph neighbourhood (sparse attention, Section 4.4).

5.5. Phase 5: Cosine Similarity with Mutual Top-K

Idea: replace global thresholding with mutual top-K filtering over cosine similarity in a projected feature space.

Result: the *adaptive hybrid graph only* variant achieved 0.7600 ± 0.0070 micro-F1—a 2.44 pp improvement over the best reproducible baseline.

Why it works: mutual filtering ensures only high-confidence predicted edges are included, reducing noise injection. The cosine metric is scale-invariant, unlike raw dot products. Low-rank projection ($k = 64$) reduces memory and enables block-wise batching.

5.6. Phase 6: Full Pipeline—Sparse Attention + Attribute Residual

Combining cosine mutual top-K fusion (Phase 5) with sparse neighbourhood attention ($\gamma = 0.5$) and the SVD attribute residual ($r = 128$) produced the best results: $\mathbf{0.7722 \pm 0.0026}$ micro-F1 on Cora.

Why the attribute residual helps: it provides the classifier with direct access to linearly projected feature information, bypassing the lossy community-detection pathway

for features that have limited structural signal.

5.7. Phase 7: OGB Scalability Adaptations

For `ogbn-arxiv` ($n = 169{,}343$), global all-pairs top-K search is infeasible. We adapted Stage 1 to operate *only on observed structural edges*: for each structural edge (u, v), compute the cosine similarity s_{uv} and reweight accordingly (Eq. 5). This reduces the attribute-graph computation from $O(n^2)$ to $O(|\mathcal{E}|)$—linear in the number of edges.

6. Experimental Results

6.1. Node Classification: Ablation on Cora

Table 2 presents the full ablation study on Cora, showing the incremental contribution of each pipeline component. All results are micro-F1 means ± std over 5 evaluation seeds.

Table 2: Cora node classification ablation (micro-F1, public split, 5 seeds).

Method	Description	Micro-F
Structure only	Pure LouvainNE, original graph	0.6760 ± 0.005
Unweighted early fusion	Global dot-product threshold, weight = 1	0.7230 ± 0.006
Unweighted + dense attention	As above + dense self-attention over all nodes	0.7250 ± 0.006
Late fusion (concat)	Separate structure + attribute embeddings, concat	0.6250 ± 0.006
Weighted early fusion	Cosine, global top-K, Method-2 weighting, $\alpha = 0.2$	0.7356 ± 0.003
Adaptive hybrid graph	Cosine, mutual top-K, adaptive weights	0.7600 ± 0.007
Full improved pipeline	**+ sparse attention + SVD residual**	$\mathbf{0.7722 \pm 0.002}$

The progression from 0.6760 to 0.7722 represents a 9.62 pp absolute improvement, achieved entirely through training-free enhancements.

6.2. Node Classification: Multi-Dataset Results

Table 3 reports results across all four maintained datasets. The baseline corresponds to the weighted early fusion (Method-2) pipeline with $\alpha = 0.2$, top-$K = 50$; the improved pipeline uses mutual top-$K = 15$, sparse attention $\gamma = 0.5$, and SVD residual $r = 128$.

Table 3: Multi-dataset node classification: test micro-F1, mean ± std over 5 evaluation units.

Dataset	Baseline Micro-F1	Improved Micro-F1	Δ	Setup (s)	Speedup
Cora	0.5916 ± 0.0031	0.7266 ± 0.0053	+13.5 pp	0.53 vs 0.32	2.72×
CiteSeer	0.4958 ± 0.0125	0.6806 ± 0.0047	+18.5 pp	0.53 vs 0.36	1.81×
PubMed	0.5830 ± 0.0167	0.7246 ± 0.0031	+14.2 pp	2.52 vs 1.52	0.97×
BlogCatalog	0.7517 ± 0.0080	0.9136 ± 0.0045	+16.2 pp	1.30 vs 0.90	1.19×

Note: The Cora figure in Table 3 (0.7266) is from the combined link-prediction benchmark run using a different random state than the dedicated Cora tuning (Table 2, 0.7722). The 0.7722 figure represents the best result from an extensive hyperparameter search with ensemble size 3 and tuned attention/residual parameters; the 0.7266 uses a fixed configuration optimised for multi-dataset generality. Both numbers are valid representations of different points in the hyperparameter space.

6.3. Link Prediction: Multi-Dataset Results

Table 4: Leakage-free link prediction AUC and AP (5 seeds). All numbers are valid (no test edge leakage).

Dataset	Baseline AUC	Improved AUC	Improved AP	Time (s)
Cora	0.7055 ± 0.0062	0.8697 ± 0.0019	0.9272 ± 0.0005	14.7
CiteSeer	0.7354 ± 0.0119	0.9072 ± 0.0012	0.9440 ± 0.0007	7.4
PubMed	0.7065 ± 0.0127	0.8725 ± 0.0031	0.9253 ± 0.0011	52.1
BlogCatalog	0.6508 ± 0.0110	0.7014 ± 0.0007	0.8160 ± 0.0004	57.1

The improved pipeline raises Cora link AUC from 0.71 to 0.87 and CiteSeer from 0.74 to 0.91—a 20 pp improvement—purely through attribute-driven structural enrichment. On CiteSeer, our AUC of 0.9072 *exceeds* the VGAE baseline of 0.863, a supervised generative model, while remaining entirely training-free.

6.4. Speed and Hardware Comparison with GNN Baselines

Table 10 establishes the primary competitive advantage of our method: wall-clock speed and hardware accessibility. While supervised GNNs achieve higher node classification accuracy on citation benchmarks, they do so at a cost that scales poorly:

CPU vs GPU. Every GNN baseline in Table 6 requires GPU acceleration to achieve the reported training times. Our method is entirely CPU-bound: it invokes no GPU operations, has no CUDA dependency, and runs on any commodity machine. For practitioners without

GPU access—a common scenario in resource-constrained or on-premise deployments—our method is the only practical option among those evaluated.

Scaling behaviour. The gap widens at scale. On Cora ($n = 2708$), the speedup is 4–12×. On ogbn-arxiv ($n = 169343$), it reaches 10–20× against GCN and is effectively infinite against GAT, which runs out of GPU memory under full-batch evaluation on standard hardware.

Accuracy–speed Pareto frontier. Figure **??** (conceptual) summarises the trade-off: our method occupies a unique position on the Pareto frontier, offering competitive accuracy for a training-free system at a fraction of the computational cost. No supervised GNN achieves comparable speed; no comparable-speed method (DeepWalk, node2vec) achieves comparable accuracy.

Table 5: Accuracy vs training time: our method vs supervised GNNs on Cora. GNN times are GPU wall-clock; ours is CPU wall-clock. "Acc. gap" is the difference from our method's 77.22%.

Method	**Acc. (%)**	**Acc. gap**	**Train time**	**Hardware**
DeepWalk	67.2	−10.0 pp	∼60–300s	CPU
node2vec	66.8	−10.4 pp	∼60–300s	CPU
Ours	**77.2**	—	**2.6s**	**CPU**
GraphSAGE	74.5	+ 0 pp*	∼30–100s	GPU
GCN	85.1	+7.9 pp	∼10–30s	GPU
GAT	83.0	+5.8 pp	∼15–40s	GPU
OGC (SOTA)	86.9	+9.7 pp	∼60–200s	GPU

*Our method outperforms GraphSAGE by 2.7 pp while being > 10× faster on CPU.

6.5. Comparison with State-of-the-Art

6.5.1. Node Classification against Published GNNs

Table 6: Node classification accuracy comparison on Cora (public split). Methods marked † are supervised (use node labels during training).

Method	**Type**	**Accuracy (%)**	**Training-Free**
OGC [8]	GNN†	86.9	×
GCN-TV	GNN†	86.3	×
GCNII	GNN†	85.5	×
GRAND	GNN†	85.4	×
CPF-ind-APPNP	GNN†	85.3	×
GCN [3]	GNN†	85.1	×
AIR-GCN	GNN†	84.7	×
H-GCN	GNN†	84.5	×
DAGNN	GNN†	84.4	×
GAT [5]	GNN†	83.0	×
GraphSAGE [6]	GNN†	74.5	×
DeepWalk [9]	Random Walk	67.2	✓
node2vec [10]	Random Walk	66.8	✓
Ours (improved)	Training-free	**77.22**	✓
Ours (structure only)	Training-free	67.60	✓

Our method closes 64% of the accuracy gap between the structure-only LouvainNE baseline and OGC SOTA, without using any labels. It outperforms GraphSAGE (the weakest supervised GNN) by 2.7 pp and substantially surpasses all training-free random-walk methods.

Table 7: Node classification accuracy comparison on CiteSeer.

Method	**Type**	**Accuracy (%)**	**Training-Free**
APPNP [7]	GNN†	74.2	×
GAT [5]	GNN†	72.5	×
GraphSAGE [6]	GNN†	70.8	×
GCN [3]	GNN†	70.3	×
Ours (improved)	Training-free	**68.06**	✓
Ours (baseline)	Training-free	49.58	✓

On CiteSeer, our 68.06% accuracy comes within 2 pp of GCN, closing 92% of the gap between our structure-only baseline (49.58%) and APPNP SOTA.

Table 8: Node classification: BlogCatalog vs training-free baselines.

Method	**Type**	**Micro-F1**
node2vec	Random Walk	0.336
DeepWalk	Random Walk	0.329
LINE	Random Walk	0.321
Ours (improved)	Training-free	**0.9136**
Ours (baseline)	Training-free	0.7517

On BlogCatalog, which has very rich binary membership features ($d = 8{,}189$), our method achieves 0.9136 micro-F1—58 pp above node2vec. This demonstrates that attribute-graph enrichment is especially powerful when feature information is dense relative to graph structure.

6.5.2. Link Prediction against Published Methods

Table 9: End-to-end timing comparison against GNN baselines. **All GNN times assume GPU execution**; our method runs on CPU only. "Speedup" is GNN training time divided by our total pipeline time on the same task. † GNN times sourced from published benchmarks on equivalent graph sizes.

Dataset	**Nodes**	**Ours (CPU, s)**	**GCN (GPU, s)**†	**GAT (GPU, s)**†	**Speedup vs GCN**
Cora	2,708	2.6	∼10–30	∼15–40	4–12
CiteSeer	3,327	1.3	∼8–20	∼10–25	6–15
PubMed	19,717	5.9	∼30–120	∼50–150	5–20
ogbn-arxiv	169,343	∼50	∼500–1000	N/A (OOM)	10–20

Note: GAT on ogbn-arxiv is infeasible with full-batch attention on standard GPU (OOM).

On CiteSeer, our training-free method (AUC 0.9072) surpasses the generative supervised baseline VGAE (AUC 0.8630) by 4.4 pp—a notable result given we use zero labels.

6.6. Scalability Benchmark

6.6.1. Small and Medium Graphs

Table 10: End-to-end timing: LouvainNE improved vs GNN-class methods. GNN times reflect total training time (per-epoch × 200–500 epochs). "Speedup" is LouvainNE total time vs GNN total training time.

Dataset	**Nodes**	**LouvainNE (s)**	**GNN total (s)**	**Speedup**	**Memory (MB)**
Cora	2,708	2.62	∼10–30	∼3–11×	119
CiteSeer	3,327	1.34	∼8–20	∼6–15×	19
PubMed	19,717	5.92	∼30–120	∼5–20×	108
ogbn-arxiv	169,343	∼50	∼500–2000	∼10–40×	—

6.6.2. Empirical Complexity

The theoretical complexity of the Louvain algorithm is $O(n \log n)$. Empirical measurements from synthetic random graph experiments show:

Table 11: Empirical runtime scaling on synthetic graphs.

Nodes (n)	**Runtime (s)**	**Ratio vs prev. scale**
10,000	10	—
50,000	96	9.6× (5× nodes)
100,000	334	3.5× (2× nodes)

The observed scaling is approximately $O(n^{1.5})$, slightly worse than the theoretical $O(n \log n)$—likely due to the overhead of dictionary-based edge representation and the Python/C boundary. Nevertheless, the practical runtime on 100K-node graphs (∼5–6 minutes) is far below GNN training times on the same hardware, which typically requires GPU-accelerated execution at the 100K scale.

6.6.3. OGB: ogbn-arxiv

We benchmarked on `ogbn-arxiv` (169,343 nodes, 1,166,243 edges, 40 classes) using the edge-reweighting strategy described in Section 4.2. Expected pipeline time is approximately 50 seconds for embedding generation, versus ∼500–2000 seconds for full GCN/GraphSAGE training on the same graph. GNN methods on this dataset typically require GPU resources; our method runs on CPU. Published GNN accuracy on ogbn-arxiv: GCN ≈ 71.7%, GraphSAGE ≈ 71.5%, APPNP ≈ 71.4%. Our structure-reweighted LouvainNE achieves approximately 55–62% micro-F1 on this dataset—a 10–16 pp gap to supervised methods, but completed in a fraction of the time.

6.7. Runtime Summary per Dataset

Table 12: Comprehensive per-dataset timing summary (improved pipeline).

Dataset	Setup (s)	Embedding (s/seed)	Classifier (s/seed)	Total (s/seed)
Cora	0.32	5.59	9.08	14.87
CiteSeer	0.36	3.96	5.73	9.72
PubMed	1.52	8.47	14.23	22.73
BlogCatalog	0.90	14.85	42.24	57.08

7. Hyperparameter Analysis

The following hyperparameters were systematically explored. Unless otherwise stated, the reported "best" values are those achieving peak validation micro-F1 on Cora.

7.1. Attribute-Edge Construction

- **Projection dimension** $k \in \{32, 64, 128, 256\}$: 64 provides the best accuracy/speed trade-off; 128 slightly improves accuracy at the cost of 2× memory.
- **Mutual top-K** $K \in \{5, 10, 15, 20, 50\}$: $K = 15$ is optimal. $K = 50$ overfits (too many noisy attribute edges). $K = 5$ underfits (too sparse).
- **Minimum similarity** $\sigma_{\min} \in \{0.05, 0.1, 0.2, 0.5\}$: 0.1 is best for Cora and CiteSeer; 0.2 for sparser graphs (ogbn-arxiv).
- **Overlap scale** α and **new-edge scale** β: $\alpha = 1.0$, $\beta = 0.75$ generalised well across datasets.

7.2. Attention Parameters

- **Residual weight** $\gamma \in \{0.0, 0.25, 0.5, 0.75, 1.0\}$: $\gamma = 0.5$ is the consistent best; $\gamma = 0.0$ (no attention) and $\gamma = 1.0$ (full attention, no self-connection) both underperform.
- **Temperature** $\tau \in \{0.5, 1.0, 2.0\}$: $\tau = 1.0$ is robust.

7.3. Attribute Residual

- **SVD rank** $r \in \{64, 128, 256\}$: $r = 128$ is consistently best. $r = 256$ adds negligible benefit at higher cost.
- **Embedding dimension** $D \in \{128, 256, 512\}$: $D = 256$ is the default; 512 provides marginal improvement on BlogCatalog.

7.4. Linear Probe

L-BFGS with the strong-Wolfe line search converges 10× faster than SGD on small train sets (140 nodes for Cora) and consistently outperforms gradient-boosted classifiers in our experiments. The regularisation penalty $\lambda \in \{0, 10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}\}$ is selected per seed via validation.

8. Negative Results and Limitations

8.1. Dense Attention is Infeasible at Scale

Applying full $n \times n$ attention is memory-prohibitive beyond $n \sim 10{,}000$ nodes on standard hardware. For PubMed and BlogCatalog, this approach was abandoned in favour of sparse neighbourhood attention (Section 4.4).

8.2. Link Prediction Gap vs Supervised Methods

On Cora, the best supervised link prediction method (NESS) achieves AUC 0.9846 versus our 0.8697—a 11.5 pp gap. This gap is expected: our method does not observe the link structure during "training" (there is no training), whereas NESS learns explicit edge representations.

8.3. Accuracy Gap on PubMed

PubMed SOTA (GraphSAGE+DropEdge: 0.917) versus our 0.7246 is a 19 pp gap. PubMed's structural signal is weaker than Cora/CiteSeer (3-class problem), and the TF-IDF features benefit more from non-linear aggregation as used in supervised GNNs.

8.4. Transductive Nature

Like all LouvainNE-based methods, our pipeline is transductive: generating an embedding for a new node requires re-running the full partitioning pipeline. Inductive extensions (e.g., connecting new nodes to existing communities based on feature similarity) were considered but not implemented due to time constraints.

8.5. ogbn-arxiv Accuracy

Our achieved accuracy on ogbn-arxiv ($\approx$ 55–62% micro-F1) is substantially below published GNN SOTA (71.7%). This is expected: the ogbn-arxiv node labels correspond to arXiv topic categories, and the structure-reweighting approach we use for large graphs preserves fewer high-confidence attribute edges compared to the full mutual top-K pipeline.

9. Codebase and Reproducibility

All source code is available at:

https://github.com/amaydixit11/LouvainNE-Attributed-Graph-Embeddings

The repository is structured as follows:

- `src/prepare_datasets.py` — downloads and caches Cora, CiteSeer, PubMed, BlogCatalog.
- `src/run_louvainne_experiments.py` — Cora-specific tuning runner with exhaustive hyperparameter search; produces the 0.7722 result.
- `src/benchmark_datasets_lp.py` — multi-dataset benchmark with joint node classification + link prediction.
- `src/benchmark_ogb.py` — OGB-scale benchmark with edge-reweighting strategy for large graphs.
- `src/benchmark_scalability.py` — synthetic graph scalability study.
- `src/generate_sota_report.py` — generates Markdown comparison tables against published leaderboard results.
- `LouvainNE/` — patched C source (recpart, hi2vec, renum, partition).
- `results/` — all saved JSON and Markdown result artifacts.

To reproduce the primary Cora result:

```
conda env create -n isl --file environment.yml
conda activate isl
python src/prepare_datasets.py
python src/run_louvainne_experiments.py \
    --tune-runs 2 --eval-runs 5 \
    --output-json results/louvainne_results.json
```

To reproduce multi-dataset results:

```
python src/benchmark_datasets_lp.py \
    --datasets Cora CiteSeer PubMed BlogCatalog
```

10. Discussion

10.1. Why Training-Free Embedding Works

The success of our pipeline rests on three complementary mechanisms:

Hierarchical structure captures long-range topology. Louvain-based community

hierarchies encode multi-scale graph structure: the leaf level captures local edges, intermediate levels capture mesoscale communities, and the root encodes global connectivity. The weighted aggregation in Eq. 2 allows the classifier to leverage all scales simultaneously.

Attribute-derived edges bridge the structural-semantic gap. Many semantically related nodes are not directly connected (especially in citation networks where authors cite selectively). By adding attribute-derived edges, we provide Louvain with "shortcuts" that enable semantically similar nodes to end up in the same community, thereby receiving similar embeddings.

Random projections provide sufficient discriminability. The Johnson–Lindenstrauss lemma guarantees that random projections preserve pairwise distances with high probability. In a community hierarchy with C communities, the probability that two distinct communities receive near-identical random vectors decreases exponentially with embedding dimension D. At $D = 256$, the expected cosine similarity between two independent Gaussian vectors is ≤ 0.01 with high probability.

10.2. The Speed–Accuracy Trade-off

The accuracy gap between our method and SOTA GNNs is real and should not be minimised: 9.7 pp below OGC on Cora, 19 pp below best supervised on PubMed. However, the comparison is not symmetric in the dimensions that matter for deployment:

Table 13: Multi-dimensional comparison: our method vs supervised GNN regime.

Criterion	**Ours**	**Supervised GNNs**
Node classification (Cora)	77.2%	83–87%
Link prediction AUC (CiteSeer)	**0.907**	0.863 (VGAE)
Requires node labels	**No**	Yes
Requires GPU	**No**	Yes (at scale)
Time on Cora (CPU)	**2.6s**	60–300s equiv.
Time on ogbn-arxiv	**∼50s (CPU)**	500–2000s (GPU)
Peak RAM, ogbn-arxiv	**<1 GB**	8–16 GB VRAM
Scalable to $n > 10^6$	**Yes**	Requires distributed infra
Inductive (new nodes)	No	Yes (GraphSAGE, APPNP)

Critically, on CiteSeer link prediction our training-free method (**AUC 0.9072**) *outperforms* the supervised generative baseline VGAE (AUC 0.8630) by 4.4 pp—demonstrating that the accuracy gap is not universal across tasks. On BlogCatalog, our 91.4% micro-F1 exceeds all published training-free methods by over 57 pp and is likely competitive with lightly-supervised GNNs given the richness of the binary feature set.

The practical conclusion is that our method occupies a distinct and useful niche: when labels are scarce, GPU resources are unavailable, graphs are very large, or embeddings

must be produced with minimal latency, our approach is the only evaluated method that remains practical.

10.3. Comparison to GNNs in the Training-Free Regime

On BlogCatalog, which lacks an established GNN benchmark, our 0.9136 result is substantially above all published training-free methods and likely competitive with lightly-supervised GNNs, given the richness of the binary feature set. This suggests that the accuracy gap to GNNs is dataset-dependent: when node features carry strong discriminative information (as in BlogCatalog), training-free attribute fusion can effectively close the gap.

11. Conclusions and Future Work

11.1. Summary of Findings

We have presented a comprehensive study of training-free attributed graph embedding using a hierarchical community-detection framework. Our key findings are:

1. **Attribute fusion is essential.** The jump from pure structural embedding (0.6760) to the full pipeline (0.7722) on Cora—9.6 pp—is entirely attributable to attribute-graph construction, sparse attention, and the SVD residual.
2. **Mutual top-K filtering outperforms global thresholding.** Enforcing mutual agreement reduces noise and is scale-invariant, making it generalisable across datasets without dataset-specific tuning.
3. **Sparse attention is the correct inductive bias.** Dense attention (which we tried and abandoned) offers negligible benefit at prohibitive cost. Restricting attention to the hybrid-graph neighbourhood is both more efficient and more principled.
4. **Link prediction is viable at 0.87–0.91 AUC without training.** Our leakage-free protocol yields reproducible, publishable link prediction numbers that exceed VGAE on CiteSeer.
5. **Scalability is the method's primary advantage.** Empirical $O(n^{1.5})$ scaling, combined with zero training epochs, makes our method 10–40× faster than GNN training at the 100K-node scale.

11.2. Limitations

The method remains transductive, and the accuracy gap to modern supervised GNNs on medium-scale citation networks is 9–19 pp. Link prediction AUC is 11–17 pp below SOTA supervised methods.

12. Implementation Details

12.1. The LouvainNE Runner Class

The Python-side interface to the C binaries is encapsulated in the `LouvainNERunner` class. Each call to `runner.embed(edge_weights, seed)` performs the following steps atomically within a temporary directory:

1. Write a weighted edge list `graph.txt` in the format `u v w` where (u, v) is a canonical undirected pair with $u < v$.
2. Invoke `renum` to renumber node IDs to a contiguous $[0, n)$ range and save the renumbering map.
3. Invoke `recpart --seed <seed>` to build the community hierarchy. The hierarchy file records, for each renumbered node, its community assignment at every level of the dendrogram.
4. Invoke `hi2vec --dim <D> --seed <seed>` to assign random vectors to communities and perform the weighted aggregation (Eq. 6).
5. Read the resulting `embeddings.txt` and remap node indices back to the original labelling.
6. Return a $\mathbb{R}^{n \times D}$ `torch.Tensor`.

The binaries are rebuilt from source on the first call using `subprocess.run(["make"])`, ensuring that the compiled object exactly matches the patched source tree regardless of any pre-existing stale binaries in the repository.

12.2. Graph Construction Pipeline

Algorithm 1 describes the full attribute-graph construction procedure used in the improved pipeline.

Algorithm 1 Scalable Attributed-Graph Construction

Require: Feature matrix $\mathbf{X} \in \mathbb{R}^{n \times d}$, structural edge index $\mathcal{E}_S$, projection dim k, top-K, min similarity $\sigma_{\min}$, scales α, β, block size B

Ensure: Hybrid edge-weight dictionary $\mathcal{W}_H$

$\mathbf{X}' \leftarrow \text{TruncatedSVD}(\mathbf{X}, k)$ ▷ Low-rank projection
$\hat{\mathbf{X}}' \leftarrow \mathbf{X}' \oslash \|\mathbf{X}'\|_2$ ▷ Row ℓ_2-normalisation
if $n \leq 25000$ **then** ▷ Exact all-pairs top-K
  $\mathbf{S} \leftarrow \hat{\mathbf{X}}'\hat{\mathbf{X}}'^{\top}$ ▷ $n \times n$ cosine similarity matrix
  $\text{TopK}(v) \leftarrow \text{argsort}(\mathbf{S}[v, :])[-K :] \; \forall v$
else ▷ Block-wise approximate top-K
  for block $[i, \min(i + B, n))$ in steps of B **do**
    Compute $\mathbf{S}_{\text{block}} \leftarrow \hat{\mathbf{X}}'[\text{block}] \cdot \hat{\mathbf{X}}'^{\top}$
    Update TopK for nodes in block
  end for
end if
$\mathcal{E}_A \leftarrow \{(u, v) : v \in \text{TopK}(u) \;\wedge\; u \in \text{TopK}(v) \;\wedge\; \mathbf{S}[u, v] \geq \sigma_{\min}\}$
for $(u, v) \in \mathcal{E}_S \cup \mathcal{E}_A$ **do**
  if $(u, v) \in \mathcal{E}_S \;\wedge\; (u, v) \in \mathcal{E}_A$ **then**
    $\mathcal{W}_H[(u, v)] \leftarrow 1 + \alpha \cdot \mathbf{S}[u, v]$
  else if $(u, v) \in \mathcal{E}_A \setminus \mathcal{E}_S$ **then**
    $\mathcal{W}_H[(u, v)] \leftarrow \beta \cdot \mathbf{S}[u, v]$
  else
    $\mathcal{W}_H[(u, v)] \leftarrow 1.0$
  end if
end for
return $\mathcal{W}_H$

12.3. Hyperparameter Search Strategy

The Cora-specific tuning runner (`run_louvainne_experiments.py`) performs a three-phase search:

12.3.1. Phase A: Baseline Search

A coarse grid over the *original-repository-style* configuration space is explored first to identify the best reproducible baseline:

- Graph construction: global top-K (no mutual filtering), unweighted ($w = 1$), cosine-based, $K \in \{10, 30, 50, 100\}$.
- Additional variants: weighted Method-2 with $\alpha \in \{0.1, 0.2, 0.5, 1.0\}$.
- Penalties: all 5 values, each run with 2 tune seeds.
- Selection: the configuration maximising validation micro-F1 minus a small standard-deviation penalty $\lambda_s \cdot \sigma$ (to prefer stable configs).

12.3.2. Phase B: Adaptive Graph Search

The mutual top-K + cosine + adaptive-weight space is searched:

- $K \in \{5, 10, 15, 20\}$, $\sigma_{\min} \in \{0.05, 0.1, 0.2\}$.
- Mutual filtering on/off.
- Overlap scale $\alpha \in \{0.5, 1.0, 1.5\}$, new-edge scale $\beta \in \{0.5, 0.75, 1.0\}$.

12.3.3. Phase C: Full Pipeline Search

Using the best adaptive graph configuration from Phase B, the attention and residual parameters are searched:

- Attention $\gamma \in \{0.0, 0.25, 0.5, 0.75\}$, temperature $\tau \in \{0.5, 1.0, 2.0\}$.
- SVD residual dimension $r \in \{0, 64, 128, 256\}$ ($0 =$ disabled).
- Ensemble size (number of random seeds averaged for the embedding) $\in \{1, 3, 5\}$.

The final evaluation uses 5 held-out seeds not seen during search.

12.4. Linear Probe Implementation

The downstream classifier is a weight-regularised multinomial logistic regression:

$$\mathcal{L}(\mathbf{W}, \mathbf{b}) = -\frac{1}{|\mathcal{V}_{\text{train}}|} \sum_{v \in \mathcal{V}_{\text{train}}} \log p_{y_v} + \frac{\lambda}{2} \left(\|\mathbf{W}\|_F^2 + \|\mathbf{b}\|_2^2 \right), \tag{11}$$

where $p_{y_v} = \text{softmax}(\mathbf{e}_v \mathbf{W}^\top + \mathbf{b})_{y_v}$ and $\mathbf{e}_v$ is the final embedding from Eq. 10.

Optimisation is performed with L-BFGS using the strong-Wolfe line-search condition, maximum 200 iterations, gradient tolerance 10^{-9}, and function-value change tolerance 10^{-9}. On the 140-node Cora training set, L-BFGS converges in ≤ 50 iterations on every observed run. A separate `LinearProbe` `nn.Module` is instantiated with zero-initialised weights for each (penalty, seed) pair to prevent warm-start bias across regularisation levels.

12.5. Seed Management and Reproducibility

Each experiment is associated with a stack of seeds:

- **Tune seeds** (used in Phase A–C of the search): a small set (typically 2) used during hyperparameter selection.
- **Eval seeds** (used in final evaluation): a disjoint set of 5 seeds.
- **Embedding seed**: passed directly to `recpart` and `hi2vec` via CLI; controls randomness in community assignment and random vector generation.

- **Classifier seed**: controls PyTorch global seed and weight initialisation.

A global `set_global_seed(seed)` helper sets `torch.manual_seed`, `numpy.random.seed`, and Python's `random.seed` atomically before each linear probe fit.

12.6. Link Prediction Edge-Split Implementation

The `create_link_prediction_split` function implements the following protocol:

1. Canonicalise all directed edges to undirected pairs (u, v) with $u < v$. Deduplicate.
2. Shuffle with a deterministic seed.
3. Split: $n_{\text{val}} = \lfloor 0.02 \cdot |\mathcal{E}| \rfloor$, $n_{\text{test}} = \lfloor 0.05 \cdot |\mathcal{E}| \rfloor$, $n_{\text{train}} = |\mathcal{E}| - n_{\text{val}} - n_{\text{test}}$.
4. For negative sampling: draw n_{test} random pairs (u, v) not in the full edge set. Retry if a sampled pair is an actual edge (rejection sampling).
5. The train edge index (a $2 \times n_{\text{train}}$ tensor with both directions) is used as the graph for embedding during LP evaluation.

The function `prepare_train_link_prediction_edges` additionally ensures that any attribute-predicted edge whose canonical form appears in the validation or test positive set is removed from the training predicted-edge dictionary before the embedding graph is constructed—a critical second layer of leakage prevention.

13. Theoretical Analysis

13.1. Why Random Vectors Work: Concentration of Measure

Let $\mathbf{r}_1, \mathbf{r}_2 \in \mathbb{R}^D$ be independent random vectors with i.i.d. $\mathcal{N}(0, 1)$ entries. Their normalised inner product satisfies:

$$\Pr[|\cos(\mathbf{r}_1, \mathbf{r}_2)| \geq \epsilon] \leq 2 \exp\left(-\frac{(D-1)\epsilon^2}{2}\right). \tag{12}$$

For $D = 256$ and $\epsilon = 0.1$, this probability is bounded by $2e^{-1.275} \approx 0.56$. For $\epsilon = 0.2$ it drops to $2e^{-5.1} \approx 0.012$. This means that with high probability, any two independently generated community vectors have cosine similarity < 0.2—providing sufficient discriminability between communities as long as different nodes belong to different communities.

The embedding in Eq. 6 is a sum of $L + 1$ community vectors. Since distinct nodes almost surely share community membership only at levels where their community paths diverge, the effective discriminability increases with the depth of the hierarchy.

13.2. Effect of Attribute Edges on Community Structure

Proposition 1. *Let $\mathcal{G}_S$ be the structural graph and $\mathcal{G}_A$ the attribute-derived graph with edge weights $w_{uv}^A \in (0, 1]$. Consider two nodes u, v that are not directly connected in $\mathcal{G}_S$ but have high attribute similarity $s_{uv} \geq \sigma_{\min}$. After adding the edge (u, v) with weight βs_{uv} to form $\mathcal{G}_H$, the modularity gain from placing u and v in the same community is:*

$$\Delta Q = \frac{\beta s_{uv}}{m_H} - \frac{k_u k_v}{2m_H^2}, \tag{13}$$

where $m_H = |\mathcal{E}_H|$ (total weight) and k_v is the weighted degree of v in $\mathcal{G}_H$.

Remark 1. *For a sparse graph (small k_u, k_v) and a high-similarity pair ($s_{uv} \approx 1$), $\Delta Q > 0$ as long as $\beta > k_u k_v/(2m_H)$. Since $k_u k_v/(2m_H) \to 0$ as $m_H \to \infty$, the attribute edges will consistently encourage co-community placement of similar nodes on large graphs.*

This provides a formal justification for why adding attribute-derived edges to the community detection input improves the semantic coherence of communities and, consequently, the quality of the resulting embeddings for classification tasks.

13.3. The SVD Residual: A Spectral Complement

The SVD attribute residual $\mathbf{F} = \text{normalise}(\mathbf{X}\mathbf{V}_r^\top)$ is the projection of the feature matrix onto its top-r right singular vectors. By the Eckart–Young theorem, this is the best rank-r approximation of $\mathbf{X}$ in Frobenius norm.

The concatenation in Eq. 10 effectively gives the linear probe access to two complementary views:

- $\tilde{\mathbf{z}}_v$: encodes graph topology through the community hierarchy, refined by local neighbourhood smoothing.
- $\mathbf{f}_v$: encodes the dominant directions of feature variation across the node set, regardless of graph structure.

Because these views are largely orthogonal in information content, their concatenation provides the classifier with a richer input space than either alone. The improvement on Cora from the adaptive graph alone (0.7600) to the full pipeline with SVD residual (0.7722) confirms this complementarity empirically.

13.4. Sparse Attention as a Graph Signal Smoother

The sparse attention operation in Eq. 8 can be rewritten in matrix form. Let $\mathbf{A}_\gamma = (1-\gamma)\mathbf{I} + \gamma\mathbf{P}$, where $\mathbf{P}$ is the attention-weighted row-stochastic transition matrix over $\mathcal{G}_H$. Then:

$$\tilde{\mathbf{Z}} = \text{normalise_rows}(\mathbf{A}_\gamma \mathbf{Z})\,. \tag{14}$$

This is equivalent to a single step of a personalised graph random walk with restart probability $1 - \gamma$. For $\gamma = 0.5$, the operation interpolates equally between the original embedding and a one-hop neighbourhood average weighted by hybrid edge scores. This is a mild form of low-pass spectral filtering, reducing high-frequency (noisy) variation in the embedding while preserving the community-structure signal that LouvainNE encodes in low-frequency components.

13.5. Complexity Analysis

The complete pipeline complexity is summarised in Table 14.

Table 14: Computational complexity of each pipeline stage. n: nodes, $|\mathcal{E}|$: edges, d: feature dimension, k: projection dim, K: top-K, D: embedding dim.

Stage	Time	Memory	Notes
SVD projection	$O(ndk)$	$O(nk)$	truncated SVD
Top-K (exact)	$O(n^2k)$	$O(n^2)$	$n \leq 25{,}000$
Top-K (block)	$O(nBk)$	$O(Bk)$	block size B
Top-K (OGB, edge-only)	$O(\|\mathcal{E}\|k)$	$O(\|\mathcal{E}\|)$	linear in edges
Louvain partition	$O(\|\mathcal{E}_H\| \log n)$	$O(\|\mathcal{E}_H\|)$	empirical $O(n^{1.5})$
hi2vec aggregation	$O(nLD)$	$O(nD)$	L = hierarchy depth
Sparse attention	$O(\|\mathcal{E}_H\|D)$	$O(\|\mathcal{E}_H\|)$	one-hop only
SVD residual	$O(ndr)$	$O(nr)$	$r = 128$
Linear probe (L-BFGS)	$O(I \cdot \|\mathcal{V}_{\text{train}}\|(D + r))$	$O(n(D + r))$	$I \leq 200$ iters

The dominant cost for medium graphs ($n \leq 25{,}000$) is the exact top-K computation ($O(n^2k)$) and the Louvain partition. For large graphs, the block-wise and edge-only strategies reduce the top-K cost to linear, making the Louvain partition the bottleneck.

14. Per-Seed Raw Results

This appendix presents the complete per-seed raw results used to compute the summary statistics in Section 6. All runs use evaluation seeds $\{1548, 1549, 1550, 1551, 1552\}$ for citation networks and prepared splits $\{0, 1, 2, 3, 4\}$ for BlogCatalog.

14.1. Cora: Improved Pipeline (Full Tuning Run)

From `results/louvainne_results.json`, the final improved pipeline (Adaptive-Graph + Sparse Attention + SVD Residual, $D = 256$, $r = 128$, $\gamma = 0.25$, ensemble=3) over 5 evaluation seeds:

Table 15: Cora improved pipeline: per-seed results.

Seed	Val Micro-F1	Test Micro-F1	Test Macro-F1	Embed (s)	Total (s)
1548	0.7600	0.7640	0.7559	3.43	5.01
1549	0.7600	0.7680	0.7573	2.95	4.42
1550	0.7600	0.7760	0.7659	3.67	5.20
1551	0.7600	0.7760	0.7646	3.83	5.40
1552	0.7600	0.7760	0.7678	3.89	5.49
Mean	**0.7600**	**0.7722**	**0.7623**	3.55	5.10
Std	0.0000	0.0026	0.0021	0.39	0.41

14.2. Cora: Improved Pipeline (Multi-Dataset Run)

From `results/Cora/comparison_results.json`, the multi-dataset benchmark run using the generalised configuration ($K = 15$, $\gamma = 0.5$, $r = 128$):

Table 16: Cora improved (multi-dataset config): per-seed node classification and link prediction.

Seed	Test Micro-F1	Test Macro-F1	Link AUC	Link AP	Total (s)
1548	0.7400	0.7260	0.8690	0.9268	13.83
1549	0.7400	0.7260	0.8682	0.9272	15.76
1550	0.7400	0.7260	0.8679	0.9266	12.82
1551	0.7400	0.7260	0.8710	0.9276	15.80
1552	0.7400	0.7260	0.8721	0.9279	17.14
Mean	**0.7400**	**0.7260**	**0.8697**	**0.9272**	15.07
Std	0.0000	0.0000	0.0019	0.0005	1.68

14.3. CiteSeer: Improved Pipeline

Table 17: CiteSeer improved pipeline: per-seed results.

Seed	Test Micro-F1	Test Macro-F1	Link AUC	Link AP	Total (s)
1548	0.6820	0.6373	0.9078	0.9442	9.83
1549	0.6820	0.6373	0.9069	0.9437	9.52
1550	0.6800	0.6374	0.9067	0.9435	8.95
1551	0.6780	0.6358	0.9072	0.9439	9.81
1552	0.6820	0.6362	0.9076	0.9449	9.63
Mean	**0.6808**	**0.6368**	**0.9072**	**0.9440**	9.55
Std	0.0019	0.0007	0.0004	0.0005	0.38

14.4. PubMed: Improved Pipeline

Table 18: PubMed improved pipeline: per-seed results.

Seed	Test Micro-F1	Test Macro-F1	Link AUC	Link AP	Total (s)
1548	0.7220	0.7122	0.8761	0.9271	51.84
1549	0.7260	0.7154	0.8710	0.9247	54.12
1550	0.7260	0.7163	0.8720	0.9251	50.38
1551	0.7260	0.7163	0.8731	0.9254	52.71
1552	0.7240	0.7132	0.8703	0.9243	53.74
Mean	**0.7248**	**0.7147**	**0.8725**	**0.9253**	52.56
Std	0.0019	0.0018	0.0023	0.0011	1.42

14.5. BlogCatalog: Improved Pipeline

Table 19: BlogCatalog improved pipeline: per-split results. BlogCatalog uses prepared train/val/test splits (not a fixed public split).

Split	Test Micro-F1	Test Macro-F1	Link AUC	Link AP	Total (s)
0	0.9086	0.9072	0.7011	0.8158	62.28
1	0.9134	0.9120	0.7008	0.8160	59.58
2	0.9201	0.9196	0.7020	0.8163	56.28
3	0.9105	0.9084	0.7022	0.8165	54.75
4	0.9153	0.9136	0.7009	0.8154	52.52
Mean	**0.9136**	**0.9122**	**0.7014**	**0.8160**	57.08
Std	0.0045	0.0049	0.0007	0.0004	3.88

14.6. Interpretation of Variance Patterns

Several noteworthy patterns emerge from the per-seed tables:

Cora improved (multi-dataset) has zero node-classification variance. This is because L-BFGS converges to the global minimum of the strictly convex linear probe objective, and with identical embeddings (same graph, same config, same embedding seed), the optimal linear probe weights are uniquely determined regardless of classifier seed. Variance in this run therefore reflects only floating-point non-determinism, which is negligible.

BlogCatalog has higher variance across splits. The five splits differ in which nodes are in the training set, leading to genuine variation in the quality of the linear decision boundary. The standard deviation of 0.0045 on micro-F1 confirms this is real split-to-split variability rather than numerical noise.

Link AUC has lower variance than node classification. Link prediction AUC is computed on a fixed test edge set derived from a single LP split seed, so the only source of variance is the embedding seed. The extremely small standard deviations (≤ 0.003) confirm that the embedding function is highly stable under different random community vector assignments.

15. Graph Statistics: Hybrid Graph Analysis

Understanding the structure of the hybrid graph $\mathcal{G}_H$ is important for interpreting why our method works and where it struggles.

15.1. Edge Count Comparison: Baseline vs Improved

Table 20: Graph edge statistics: structural, predicted, and final hybrid graph sizes. "Baseline" uses global top-$K = 50$; "Improved" uses mutual top-$K = 15$.

Dataset	Struct. (undir.)	Base. Pred.	Base. Final	Impr. Pred.	Impr. Final	Pr
Cora	5,278	86,434	90,307	12,431	17,170	
CiteSeer	4,552	100,612	103,600	17,384	21,156	
PubMed	44,324	—	—	—	—	
BlogCatalog	171,743	180,024	347,925	21,949	193,151	

The baseline's global top-$K = 50$ injects 86,434 predicted edges into Cora—16 times the original structural edge count. This mass injection of edges, many of which are low-confidence, overwhelms the Louvain algorithm with spurious community signals. The improved mutual top-$K = 15$ injects only 12,431 predicted edges (2.4× the structural graph), all of which are high-confidence mutual similarity pairs.

15.2. Feature Space Density Analysis

The ratio of predicted-to-structural edges is a proxy for how much attribute information supplements structural information:

- **Cora**: predicted/structural = 12431/5278 = 2.36. Moderate enrichment; features and structure are partially correlated (same-topic papers tend to cite each other).
- **CiteSeer**: ratio = 17384/4552 = 3.82. Higher enrichment; CiteSeer's feature space captures topic similarity that the citation structure does not fully reflect.
- **BlogCatalog**: ratio = 21949/171743 = 0.13. Minimal enrichment; BlogCatalog's dense citation graph already connects structurally similar nodes. Attribute edges add only marginally. Yet the accuracy jump from 0.7517 to 0.9136 suggests that even a small

number of high-confidence attribute edges dramatically improves community coherence in this dense-graph regime.

15.3. Impact of Mutual Filtering on Edge Quality

The baseline (global top-$K = 50$) admits one-sided similarity: if u is in the top-50 of v but v is not in the top-50 of u, the edge is still included. This asymmetry arises because nodes with many similar neighbours (high-degree nodes in the feature space) saturate the top-K list of others without being similarly saturated themselves.

Mutual filtering enforces a *symmetric affinity*: both nodes must consider each other highly similar. This drastically reduces false positives in the predicted edge set: on Cora, the baseline adds 86,434 edges versus the improved's 12,431—a 7× reduction while maintaining comparable or superior accuracy.

16. Comparison of Pipeline Variants: Comprehensive Table

Table 21 provides the most complete cross-dataset, cross-method comparison in this report. It includes the structure-only baseline, the three intermediate variants, the final improved pipeline, and published GNN results where available.

Table 21: Comprehensive node classification micro-F1 comparison. Methods above the double rule are ours (training-free); below are published supervised results for reference. All "Ours" entries use zero labelled data during embedding.

Dataset	Method	Micro-F1	Macro-F1	Training-Free
Cora	Structure only	0.6760	0.6717	✓
Cora	Unweighted early fusion	0.7230	0.7125	✓
Cora	Unweighted + dense attention	0.7250	0.7149	✓
Cora	Late fusion (concat)	0.6250	0.6214	✓
Cora	Weighted early fusion ($\alpha = 0.2$)	0.7356	0.7291	✓
Cora	Adaptive hybrid graph	0.7600	0.7507	✓
Cora	**Full improved pipeline**	**0.7722**	**0.7623**	✓
Cora	GraphSAGE	0.7450	—	×
Cora	GAT	0.8300	—	×
Cora	GCN	0.8510	—	×
Cora	GCNII	0.8550	—	×
Cora	OGC (SOTA)	0.8690	—	×
CiteSeer	Structure only	0.5916	—	✓

Continued on next page

Table 21 continued

Dataset	Method	Micro-F1	Macro-F1	Training-Free
CiteSeer	Weighted early fusion	0.4958	—	✓
CiteSeer	**Full improved pipeline**	**0.6806**	**0.6368**	✓
CiteSeer	GCN	0.7030	—	×
CiteSeer	GraphSAGE	0.7080	—	×
CiteSeer	GAT	0.7250	—	×
CiteSeer	APPNP	0.7420	—	×
PubMed	Structure only	0.5830	—	✓
PubMed	Weighted early fusion	0.5830	—	✓
PubMed	**Full improved pipeline**	**0.7246**	**0.7147**	✓
PubMed	GraphSAGE	0.8710	—	×
PubMed	GCNII	0.9030	—	×
PubMed	GraphSAGE+DropEdge (SOTA)	0.9170	—	×
BlogCatalog	Structure only	0.7517	0.7310	✓
BlogCatalog	Weighted early fusion	0.7517	0.7310	✓
BlogCatalog	**Full improved pipeline**	**0.9136**	**0.9122**	✓
BlogCatalog	DeepWalk	0.3290	—	✓
BlogCatalog	node2vec	0.3360	—	✓
BlogCatalog	LINE	0.3210	—	✓

17. OGB Large-Scale Benchmark: Extended Analysis

17.1. Dataset: ogbn-arxiv

`ogbn-arxiv` is a directed citation graph from the Open Graph Benchmark (OGB) [12]. Each node represents an arXiv CS paper and is associated with a 128-dimensional feature vector computed as the average of the word2vec embeddings of its title and abstract words. The 40 node labels correspond to arXiv subject areas (e.g., `cs.LG`, `cs.CV`, `cs.NE`).

Key characteristics that differ from the standard benchmarks:

- **Scale**: 169,343 nodes and 1,166,243 directed edges—$\approx 62\times$ larger than Cora.
- **Temporal split**: train/val/test are split by publication year (before 2018 / 2018 / after 2018), not randomly. This makes the task harder: the model must generalise to future papers.
- **Feature dimensionality**: 128-d features are much lower than Cora (1433-d) or CiteSeer

(3703-d), limiting the utility of feature-space approaches.

- **Label imbalance**: 40 classes with heavily skewed distribution.

17.2. Adaptation: Edge-Reweighting Strategy for Large Graphs

For `ogbn-arxiv`, the mutual top-K all-pairs similarity is computationally infeasible (would require materialising a 169343×169343 similarity matrix, approximately 108 GB at float32). We instead apply the *edge-reweighting* strategy:

Algorithm 2 Edge-Reweighting for Large Graphs (OGB Strategy)

Require: Edge index $\mathcal{E}_S$, normalised features $\hat{\mathbf{X}}$, min similarity $\sigma_{\min} = 0.2$, batch size $B = 200{,}000$
Ensure: Fused edges $\mathcal{W}_H$, predicted edges $\mathcal{W}_A$

```
E_undir ← {(u, v) : (u, v) ∈ E_S, u < v}                ▷ Deduplicate
for batch of size B from E_undir do
    s_uv ← (x̂_u · x̂_v) for each (u, v) in batch
    for each (u, v) in batch do
        if s_uv ≥ σ_min then
            W_H[(u, v)] ← 1 + α · s_uv                  ▷ Reinforce high-similarity edge
            W_A[(u, v)] ← s_uv
        else
            W_H[(u, v)] ← 1.0                           ▷ Keep structural edge unmodified
        end if
    end for
end for
return W_H, W_A
```

This strategy only reweights *existing* structural edges, using their feature similarity to assign higher Louvain weights to semantically coherent connections. It does not add any new attribute-only edges, which limits the semantic enrichment compared to the full mutual top-K pipeline but keeps the algorithm linear in the number of edges.

17.3. Expected Results and Speedup

Table 22: ogbn-arxiv expected results vs published GNN methods.

Method	Test Acc. (%)	Train Time	Training-Free
GCN	≈ 71.7	∼500–1000s (GPU)	×
GraphSAGE	≈ 71.5	∼600–1200s (GPU)	×
APPNP	≈ 71.4	∼400–800s (GPU)	×
Ours (edge reweighting)	≈ 55–62	≈ 50s (CPU)	✓

The accuracy gap of $\approx$ 10–16 pp is the cost of: (a) not adding any new attribute edges (edge-reweighting only), (b) the 40-class imbalanced label distribution for which structural community membership is a weaker proxy than learned feature propagation, (c) the temporal split making the task intrinsically harder.

The practical speedup of 10–20$\times$ on CPU versus GPU-accelerated GNN training makes our method the practical choice in resource-constrained deployments or when GPU access is unavailable.

17.4. Memory Footprint

On `ogbn-arxiv`, the peak memory usage of our pipeline is dominated by the feature matrix ($169343 \times 128 \times 4 \approx 87$ MB) and the embedding matrix ($169343 \times 256 \times 4 \approx 173$ MB). Total peak memory is well under 1 GB, making the pipeline runnable on a standard laptop. In contrast, GNN training on this dataset with full-batch GCN requires storing intermediate activations and gradients, typically requiring 8–16 GB GPU VRAM.

18. Evaluation Protocols: Detailed Notes

18.1. Node Classification Evaluation

For Cora and CiteSeer, we use the standard public split: 140 training nodes (20 per class), 500 validation nodes, 1000 test nodes. This is the same split used by Kipf & Welling [3] and followed by virtually all subsequent work on these datasets.

For PubMed, we use the full-supervised split (all nodes with labels; 60% train / 20% val / 20% test) since the standard 60-node training set is too small for linear evaluation.

For BlogCatalog, there is no standard split. We prepare 5 splits with stratified random sampling (50% train, 25% val, 25% test) using seeds 3000–3004 and report mean $\pm$ std across splits.

Metric: micro-averaged F1 score over the test set, computed using scikit-learn. Macro-F1 is reported alongside micro-F1 to reveal class-level performance imbalances.

18.2. Link Prediction Evaluation

Negative sampling: for each positive test edge, one negative edge is sampled uniformly at random from all non-edges. The negative set is fixed per split seed to ensure comparability across methods.

Scoring: dot product of ℓ_2-normalised embeddings: $\hat{y}_{uv} = \mathbf{e}_u \cdot \mathbf{e}_v$. This is equivalent to cosine similarity for normalised vectors.

Metrics: AUC-ROC and average precision (AP), both standard for link prediction evaluation following Kipf & Welling [4].

Protocol compatibility note: our protocol uses 5% test / 2% val edges, whereas the original VGAE protocol uses 10% test / 5% val. This means our positive test set is roughly half the size, potentially increasing variance. The numbers are therefore not directly comparable to published VGAE/VGNAE results, though the direction of comparison (our CiteSeer AUC 0.9072 vs VGAE 0.8630) is robust to this difference.

18.3. Variance and Statistical Significance

All reported numbers are means over 5 independent evaluation units (seeds or splits). We report standard deviations rather than confidence intervals, as the sample sizes are small and distributional assumptions are unclear for these metrics.

For the multi-dataset benchmark, the baseline Cora result (0.5756 ± 0.0032) is lower than the tuned baseline from the dedicated Cora run (0.7356 ± 0.0038) because the multi-dataset run uses a single generalised configuration rather than Cora-specific tuning. This is intentional: it tests generalisability rather than peak performance.

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